SWARRNIM INSTITUTE OF TECHNOLOGY

B.Tech 1st SEMESTER — PYTHON PROGRAMMING MOCK TEST (SET 2)

Practice Paper | Similar Pattern and Difficulty, Different Questions

Instructions:

Attempt all questions in Q.1.

In Q.2, Q.3 and Q.4, attempt any TWO sub-questions from each section.

Each question in Q.1 (A) and Q.1 (B) carries 1 mark. Each attempted sub-question in Q.2, Q.3 and Q.4 carries 5 marks.

Use any appropriate Python approach. No specific loop or conditional method is required unless stated in a question.

Q.1 (A) Multiple Choice Questions (MCQs) — 05 Marks

Attempt all questions. Each question carries 1 mark.

1. What is the value of the expression?
print(20 - 3 * 4 + 2)

A. 10 B. 8

C. 6 D. 70

2. What is printed?
print(not (4 < 2) and (6 != 6))

A. True B. False

C. 0 D. 6

3. What is the output?
name = "computer"
print(name[::3])

A. cpr B. cmt C. cte D. cou

C. cte D. cou

4. How many times does the loop body run?
for k in range(3, 12, 4):
print(k)

A. 2 times B. 3 times

C. 4 times D. 9 times

5. What is printed?
for n in range(1, 5):
if n == 3:
break
print(n, end=' ')

A. 1 2 B. 1 2 3

C. 1 2 3 4 D. 3 4

Q.1 (B) Short Questions (Define / Explain) — 05 Marks

1. What is the difference between int() and float() type casting?

2. What does the split() method return when applied to a sentence?

3. What is Unicode, and how can ord() be used with a character?

4. Explain the difference between break and continue in one or two sentences.

5. What is an accumulator variable? Give a simple example of its use.

Q.2 — Attempt Any TWO — 10 Marks

(A) Explain arithmetic operators and augmented assignment operators (+=, -=, *=) with suitable examples.

(B) Explain how break, continue and pass differ. Give a short example or use case for each.

(C) Explain nested loops. Write a small program using nested loops to print every pair (i, j) where i ranges from 1 to 3 and j ranges from 1 to 2.

Q.3 — Attempt Any TWO — 10 Marks

(A) Ticket Price Write a program that accepts a person's age and prints the ticket price: below 5 → Free; 5 to 17 → ■50; 18 to 59 → ■100; 60 or above → ■70.
Test cases: 4 → Free; 12 → ■50; 65 → ■70.

(B) Perfect Number Check A number is perfect if the sum of its positive divisors excluding itself equals the number. Write a program to check whether an input number is perfect.
Test cases: 6 → Perfect; 10 → Not Perfect; 28 → Perfect.

(C) Pattern Printing Write a program that accepts N and prints this pattern.
Input: 4
Expected Output:
1
22
333
4444

Q.4 — Attempt Any TWO — 10 Marks

(A) Character Frequency Without count() Write a program that accepts a string and prints the frequency of each distinct character in the order it first appears. Do not use count() or a dictionary.
Test case: banana → b: 1, a: 3, n: 2.

(B) Alternating Sum Write a program that accepts N and calculates 1 - 2 + 3 - 4 + ... up to N.
Test cases: N = 5 → 3; N = 6 → -3.

(C) Factorial of a Number Write a program that accepts a non-negative integer and calculates its factorial using a loop.
Test cases: 5 → 120; 0 → 1; 4 → 24.

— End of Mock Test —